

Derivations and multiple factors

Given a polynomial $P(X)$ in the ring $R[X]$, where R is a (commutative) ring with identity, we define the “standard derivation” as the map $D : R[X] \rightarrow R[X]$ given by:

$$D : \sum_{k=0}^d a_k X^k \mapsto \sum_{k=1}^d (k a_k) X^{k-1}$$

This has the following properties:

- $D(a + b) = D(a) + D(b)$;
- $D(a) = 0$ for $a \in R$;
- $D(PQ) = D(P)Q + PD(Q)$;
- $D(X) = 1$.

Conversely, these properties can be seen as *uniquely* determining the map D . (More general R -derivations are those that satisfy the first three properties.)

If $P(X) = Q(X)^2 R(X)$, then we calculate

$$D(P(X)) = D(Q(X))Q(X)R(X) + Q(X)D(Q(X))R(X) + Q(X)^2 D(R(X)) = Q(X)R_1(X)$$

(This uses commutativity of R .)

In particular, when R is $\mathbb{Z}$ or a field, we have proved:

If $P(X)$ has a square factor, then $\gcd(P(X), D(P(X))) \neq 1$.

Equivalently, if R is a field and $\gcd(P(X), D(P(X))) = 1$, then the irreducible factors of $P(X)$ are distinct.